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EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

H19 Crafting Your Codespace

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Level: Introductory/Intermediate

Session Survey

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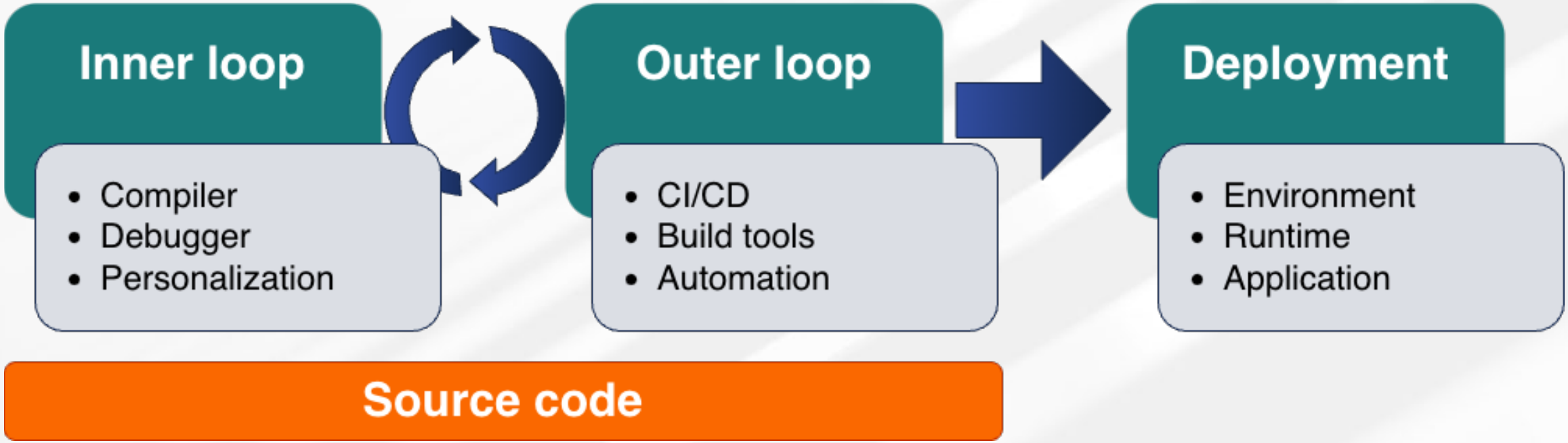
- Thank you!

Today's journey

- What are dev containers and what do they do?
- Building and configuring codespaces
- Enhancements and personalization
- Securing your workspaces

Development process



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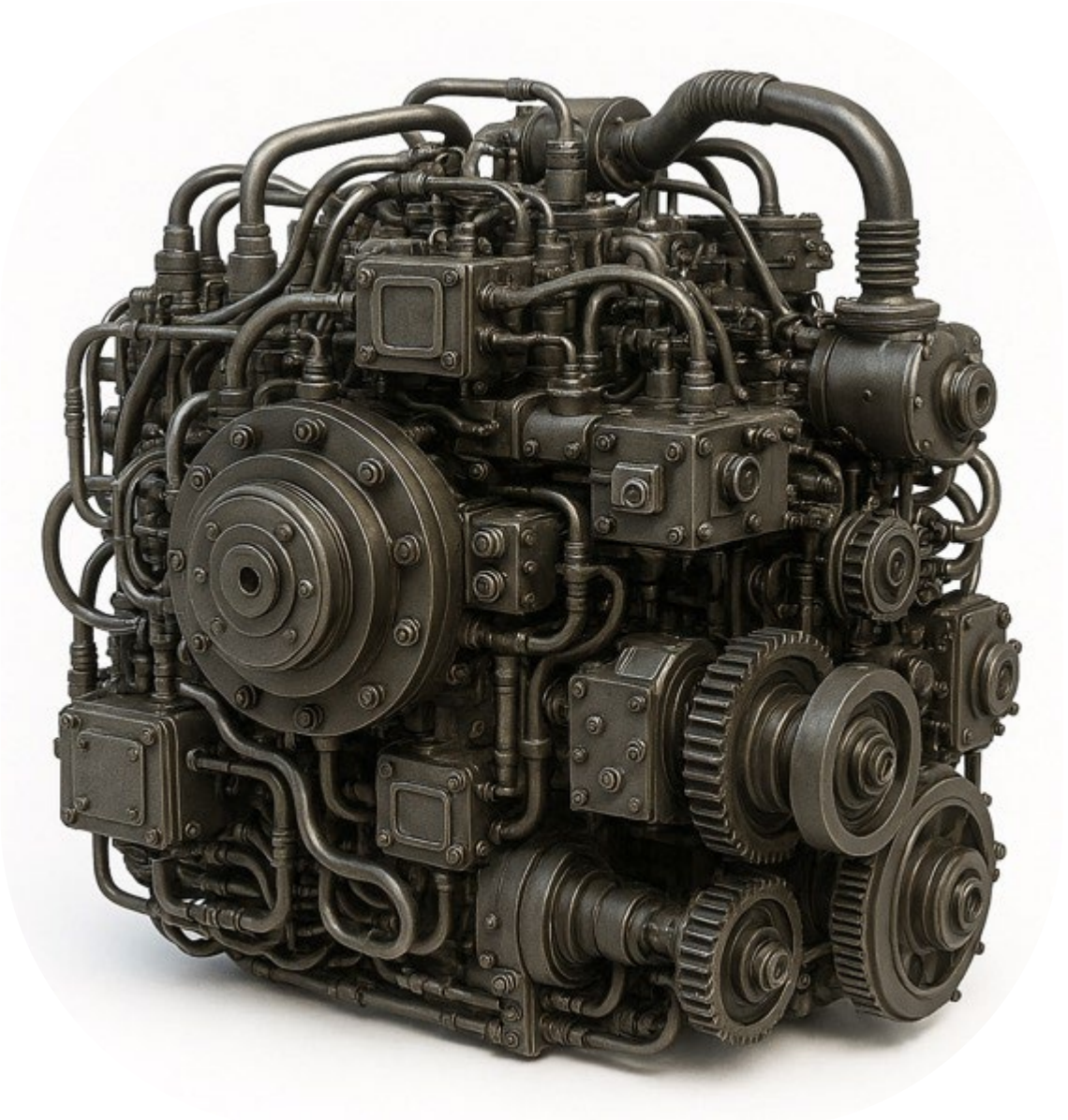
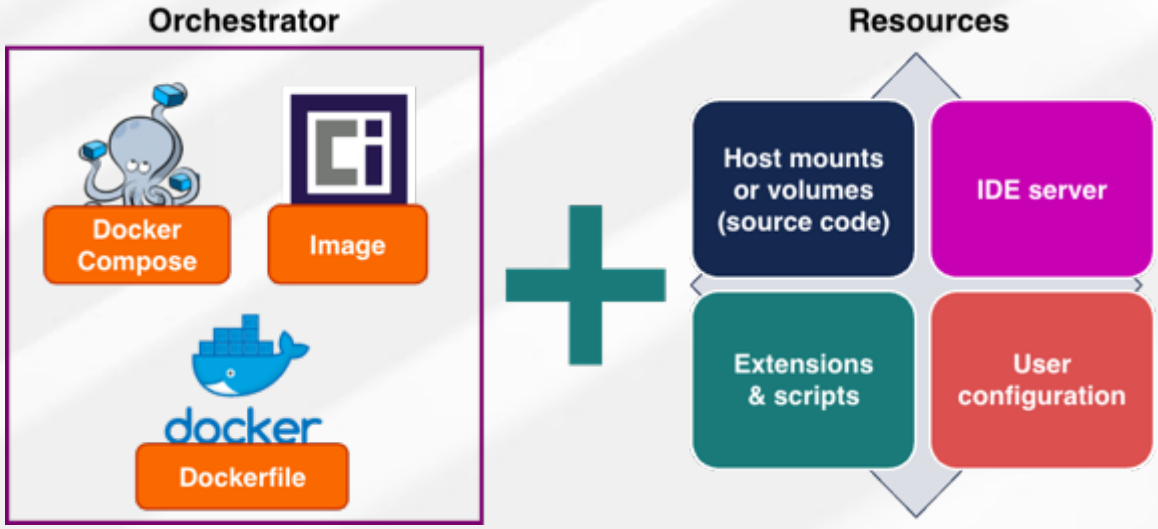
And now ... Let’s craft!

How It Works



Defining a dev container

- Isolated development environment in container
- Provides IaC for IDE and workspace settings
- Allows for customization and personalization
- Multiple containers for complex workloads



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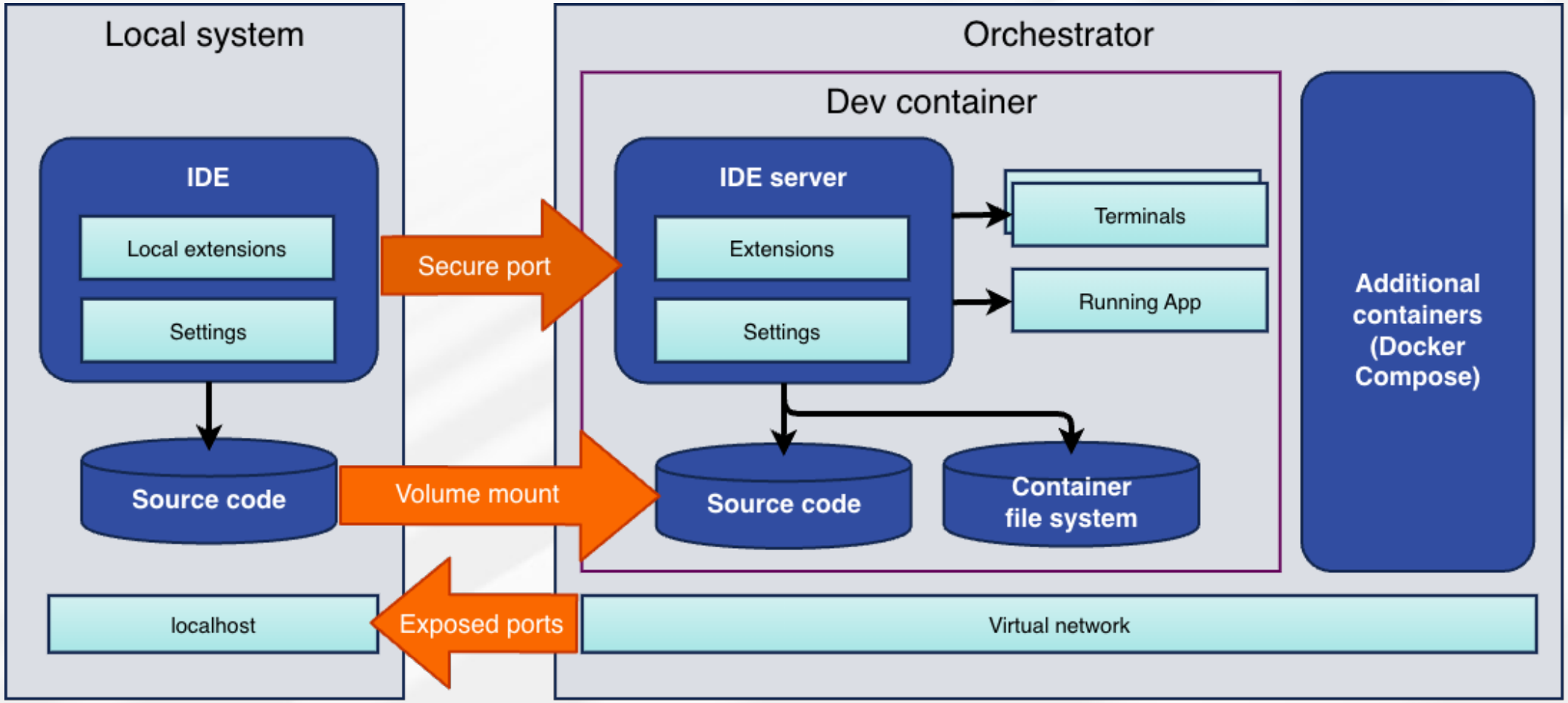
Development containers



What is GitHub Codespaces?

- GitHub's implementation of dev containers
- Server component is on VM hosted in Azure
- Automatically provides credentials for GitHub repository
- Optional browser-based VS Code or JupyterLab editor
- 120 core-hours for personal use each month (GitHub Free)

The metadata file

`.devcontainer/devcontainer.json`

- Container definition
- Tool customizations
- Host and network definitions
- Lifecycle scripts for customization
- Features

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Lifecycle

Environment creation

- `initializeCommand`
- Create image and build container

Post-creation

- `onCreateCommand`
- `updateContentCommand*`
- `postCreateCommand`

Runtime execution

- Tool configuration, including customizations & extensions
- `postStartCommand`
- `postAttachCommand`

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Personalization

- VS Code
  - Sync settings
  - Profiles
- Environment variables
- Mounts
- Dotfiles

Dotfiles

- Git repository with custom settings and configurations
- Automatically symlink all dot-prefixed files and folders
- Runs scripts to configure environment (if available)
  - `install(.sh)`
  - `bootstrap(.sh)`
  - `setup(.sh)`

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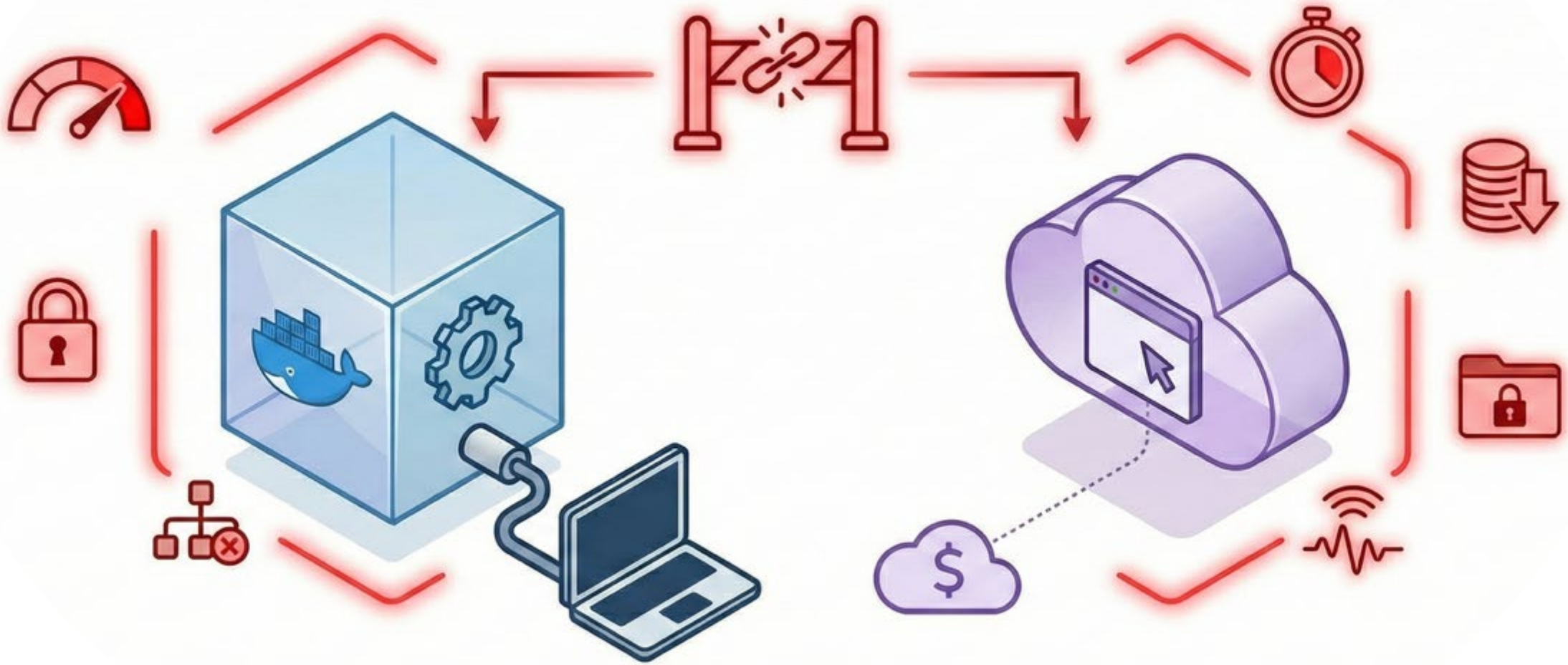
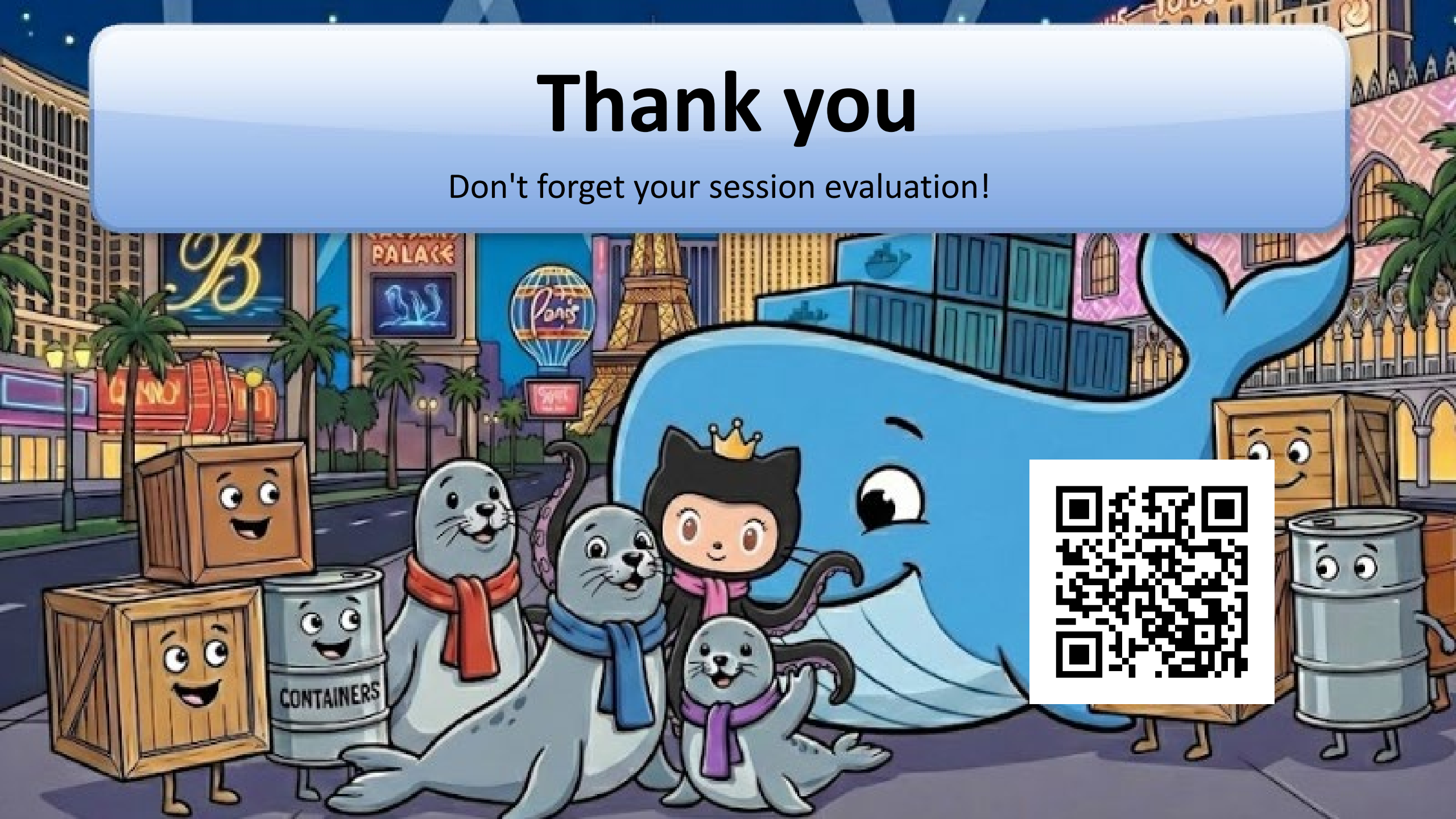
What Are The Limits?



Security

Thank you

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